

Popular Arduino Actuators & Output Modules: Servos, Buzzers, Relays, LCD, Motors

Document: EDU-ACTUATOR-01 | Comprehensive Pinout, Wiring & Specification Reference

1. SG90 Micro Servo Motor

Operating Voltage: 4.8V to 6.0V. Stall Torque: 1.8 kg-cm at 4.8V. Rotation Range: 0 to 180 degrees (standard hobby servo). Speed: 0.1 seconds per 60 degrees at 4.8V. Weight: 9 grams. Gear Material: Nylon (plastic gears). Wire Colors: Brown = GND, Red = VCC (+5V), Orange = PWM Signal.

Arduino Wiring: Brown wire to GND. Red wire to external 5V supply (NOT the Arduino 5V pin if driving multiple servos). Orange wire to an Arduino PWM pin (pins 3, 5, 6, 9, 10, or 11 on Arduino Uno). IMPORTANT: A single SG90 can draw 500-700mA under load. The Arduino USB port provides only 500mA total. Multiple servos MUST use an external 5V 2A+ power supply with common ground to Arduino.

Arduino Code: `#include <Servo.h> Servo myServo; myServo.attach(9); myServo.write(90); // Move to 90 degrees center position. myServo.write(0); // Move to 0 degrees. myServo.write(180); // Move to 180 degrees. Use map() to convert potentiometer reading to servo angle.`

2. Piezo Buzzer (Active & Passive)

Active Buzzer: Has a built-in oscillator circuit. Produces a fixed-frequency tone when DC voltage (3-5V) is applied. Wiring: Positive leg (longer, marked +) to Arduino digital pin, negative leg (shorter) to GND. Use `digitalWrite(pin, HIGH)` to sound and `digitalWrite(pin, LOW)` to silence.

Passive Buzzer: No internal oscillator. Requires an AC signal (square wave) to produce sound. Can play different frequencies and melodies. Wiring: Same as active buzzer. Use `tone(pin, frequency, duration)` to play specific notes. Example: `tone(8, 440, 500)` plays 440Hz (note A4) for 500ms. Use `noTone(pin)` to stop. Can play melodies by sequencing different frequencies.

3. 5V Relay Module (1-Channel / 2-Channel / 4-Channel)

Operating Voltage: 5V (control side). Switching Capacity: 10A at 250VAC or 10A at 30VDC (high-power side). Trigger: Active LOW (relay activates when signal pin is pulled to GND/LOW on most modules). Optocoupler Isolation: Most relay modules include an optocoupler (PC817) for electrical isolation between the Arduino logic side and the high-voltage switching side.

Pinout (Control Side): VCC = 5V, GND = Ground, IN1 = Signal input (connect to Arduino digital pin). Pinout (Switching Side): COM = Common terminal, NO = Normally Open (disconnected when relay is off, connected when relay is on), NC = Normally Closed (connected when relay is off, disconnected when relay is on). SAFETY WARNING: The high-voltage side can carry lethal mains voltage (120V/240V AC). Never touch exposed terminals when mains power is connected.

Arduino Code: `pinMode(relayPin, OUTPUT); digitalWrite(relayPin, LOW); // Relay ON (active LOW). digitalWrite(relayPin, HIGH); // Relay OFF. Use relay to switch lamps, fans, water pumps, solenoid valves, or any AC/DC device that exceeds Arduino GPIO current limits.`

4. 16x2 LCD Display (HD44780 with I2C Backpack)

Display: 16 characters per line, 2 lines. Controller: Hitachi HD44780 compatible. Backlight: Blue or green LED backlight. Operating Voltage: 5V. With I2C backpack (PCF8574): Only 4 wires needed instead of 12+ wires in parallel mode.

I2C Wiring (4 wires only): GND to GND, VCC to 5V, SDA to Arduino A4 (or SDA pin), SCL to Arduino A5 (or SCL pin). I2C Address: Usually 0x27 or 0x3F (run I2C scanner sketch to find it). Contrast: Adjustable via onboard potentiometer on the I2C backpack. If display shows only blocks, adjust contrast potentiometer.

Arduino Library: `LiquidCrystal_I2C`. **Code:**

```
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
lcd.init();
lcd.backlight();
lcd.setCursor(0, 0);
lcd.print("Hello World!");
lcd.setCursor(0, 1);
lcd.print("Line 2 text");
```

5. RGB LED (Common Cathode / Common Anode)

An RGB LED contains three separate LED dies (Red, Green, Blue) in a single 4-pin package. Common Cathode: The longest pin is the shared ground (cathode). Apply HIGH to R/G/B pins to light each color. Common Anode: The longest pin is the shared VCC. Apply LOW to R/G/B pins to light each color (inverted logic).

Wiring (Common Cathode): Red pin through 220 ohm resistor to Arduino PWM pin. Green pin through 220 ohm resistor to Arduino PWM pin. Blue pin through 220 ohm resistor to Arduino PWM pin. Longest pin (cathode) to GND. Each color needs its own current-limiting resistor.

Color Mixing with analogWrite:

```
analogWrite(redPin, 255); analogWrite(greenPin, 0); analogWrite(bluePin, 0);
// Pure Red. analogWrite(redPin, 255); analogWrite(greenPin, 255); analogWrite(bluePin, 0); // Yellow.
analogWrite(redPin, 0); analogWrite(greenPin, 0); analogWrite(bluePin, 255); // Pure Blue. Values range
0-255 for each channel (8-bit PWM).
```

6. DC Motor with L293D / L298N Motor Driver

L293D: Dual H-bridge motor driver IC. Can drive 2 DC motors simultaneously. Operating Voltage: 4.5V to 36V (motor supply). Logic Voltage: 5V. Maximum current per channel: 600mA (1.2A peak). Built-in flyback protection diodes. Pinout: Enable pins (ENA, ENB) for PWM speed control, Input pins (IN1-IN4) for direction control, Output pins (OUT1-OUT4) to motor terminals.

L298N Module: Higher current capacity (2A continuous per channel). Motor supply voltage: 5V to 35V. Has onboard 5V regulator (can power Arduino from motor supply if jumper is in place). Direction: Setting IN1=HIGH, IN2=LOW spins motor clockwise. Setting IN1=LOW, IN2=HIGH spins motor counter-clockwise. Setting IN1=IN2=LOW stops the motor (coast). Setting IN1=IN2=HIGH brakes the motor.

Speed Control: Connect ENA to an Arduino PWM pin.

```
analogWrite(enaPin, 200);
```

 sets motor speed to ~78% (200/255). **Direction:**

```
digitalWrite(in1, HIGH); digitalWrite(in2, LOW);
```

 // Forward.

```
digitalWrite(in1, LOW); digitalWrite(in2, HIGH);
```

 // Reverse. ALWAYS connect motor supply GND to Arduino GND (common ground rule).